# MILITARY SPECIFICATION
# CIRCUIT CARD ASSEMBLY: 11745246
# POSITION-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-G-48411 comprise the complete requirements for Position-3.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-G-48411 | Circuit Card Assemblies, General Specification for |

DRAWINGS

| | |
|---|---|
| 11745246 | Circuit Card Assembly, Position-3 |

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Item definition.- This assembly is one (1) of four (4) single ended printed circuit board consisting of a network of operational amplifiers, comparators, inverters, multivibrator, and gates necessary to accept position information from the gun; compare it with the position of the target and feed the results into the Visual Image Generator.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11745246.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +5 VDC
b. +15 VDC
c. -15 VDC
d. +15 VDC (Variable)

3.2.1.2 Clear Line Test.- The output on Pin 28 shall be logical "0".

3.2.2.3 Ramp Calibration.- The output from the ramp circuits shall be as shown in Figures 3 through 5.

3.2.2.4 Ramp Linearity Test.- The ramp output shall be linear in voltage and frequency for varying positive and negative voltages.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-G-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11745246 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-G-48411, Drawing 11745246 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-G-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-G-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-G-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-G-48411.



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4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-G-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-G-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power supply (PS1).- The power supply shall be 10 ± 0.5 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power supply (PS2 & PS3).- The power supply shall be 20 ± 1.0 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Power supply (variable)(PS4).- The power supply shall be adjustable with an output of ± 1 mV for any setting of from 0 to 10 VDC, 100 mA or greater, a ripple of less than 500 μVDC and a load effect of 0.01% + 1 mV.

4.2.1.4 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanosecondsor less, sweep period of 50 nanosecondsper centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.5 Dummy loads.- The dummy loads shall be one 3-input positive "Nand" gate or the equivalent to SN54L10.

4.2.1.6 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +5 ± 0.05 VDC
b. +15 ± 0.05 VDC
c. -15 ± 0.05 VDC
d. +15 ± 0.001 VDC (Variable)

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. the "1" or true output level shall be between +2.40 and +5.00 VDC.

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4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Clear Line Test.- With the external power supplies connected as shown in Figure 1, the output on pin 28 shall read logical "0".

4.2.2.5 Ramp Calibration.- The ramp output shall be as shown in Figure 2 through 5 when the procedure as outlined below as follows:

a. Apply 0.000 V ± 1 mV to pin 34.
b. Adjust R 32 until the output at pin 1 is 0.00 V ± 1 mV.
c. Increase voltage on pin 34 until -1.000 ± 1 mV is obtained on pin 1.
d. Adjust R 27 until the waveform in Figure 2 is obtained on pin 14.
e. Decrease the voltage on pin 34 until +1.000 V ± 1 mV is obtained on pin 1.
f. Adjust R 26 until the waveform in Figure 3 is obtained on pin 14.
g. Connect scope and observe waveform 4 on pin 19.
h. Apply 0.000 V ± 1 mV to pin 41.
i. Apply 10.000 V ± 1 mV to pin 37.
j. Read -2.720 V ± 30 mV on pin 32.
k. Apply 10.000 V ± 1 mV to pin 41.
l. Apply 0.000 V to pin 37.
m. Read +2.72 ± 30 mV on pin 32.
n. Reduce the voltage to 0.000 V ± 1 mV on pin 41.
o. Connect scope and observe that pin 18 is at logical "1".
p. Increase the voltage on pin 41 and observe that waveform 4 will appear when the voltage on pin 41 is 0.368 V ± 30 mV.
q. Apply logical "0" to pin 12 and waveform 4 will switch to logical "1".
r. Apply logical "1" to pin 12.
s. Decrease the voltage on pin 41 and observe that logic "1" on pin 20 shall switch to waveform 5 when the voltage on pin 41 is -0.368 V ± 30 mV.
t. Apply logical "0" to pin 12 and waveform 4 on pin 20 shall switch to logical "1".

4.2.2.6 Ramp Linearity Test.- When a voltage of from 0.00 to +10.00 is applied to pin 34 such that the voltages on pin 1 are as in column A below the corresponding frequency on pin 19 shall be as shown in column B.

| Column A | Column B |
|---|---|
| 0.100 V ± 1 mV | 6.800 ± 0.025 Hz |
| 0.500 V ± 1 mV | 34.00 ± 0.05 Hz |
| 1.000 V ± 1 mV | 68.0 ± 0.1 Hz |
| 2.000 V ± 1 mV | 136.0 ± 0.2 Hz |
| 4.000 V ± 1 mV | 272.0 ± 0.4 Hz |
| 8.000 V ± 1 mV | 544.0 ± 0.8 Hz |
| 10.000 V ± 1 mV | 680.0 ± 1.0 Hz |



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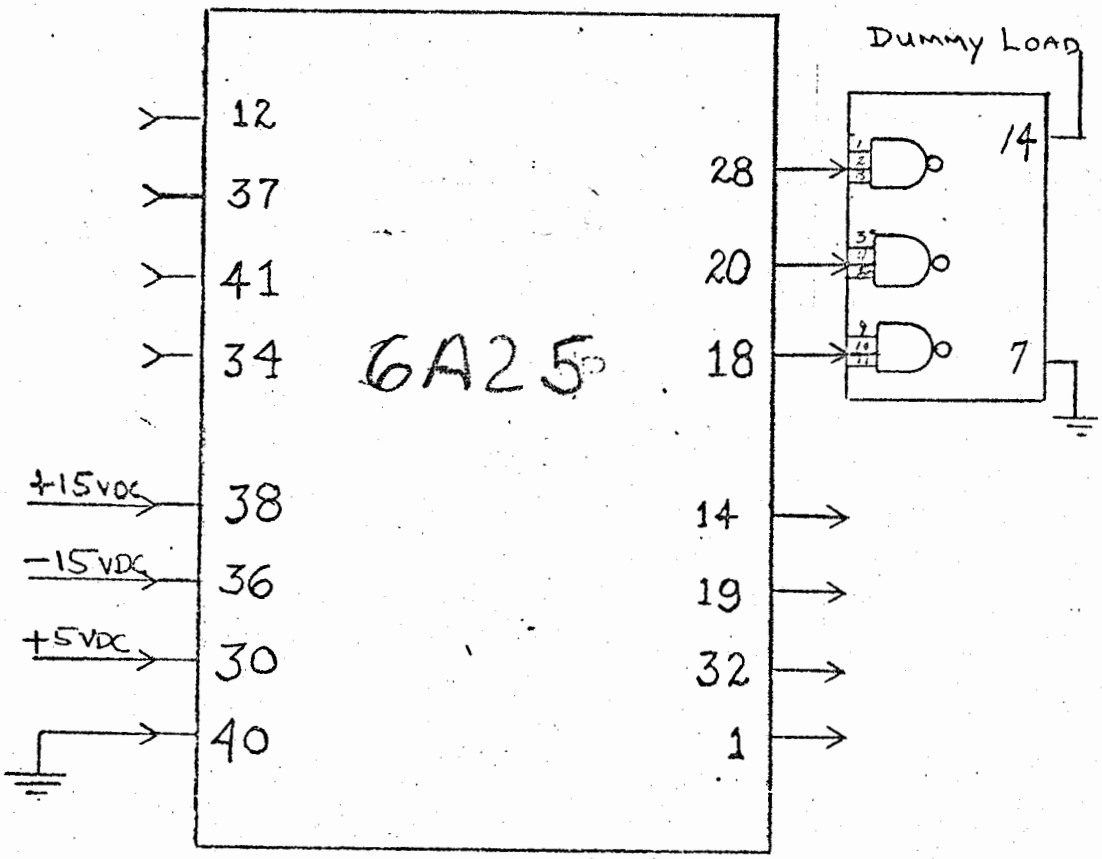



FIGURE 1

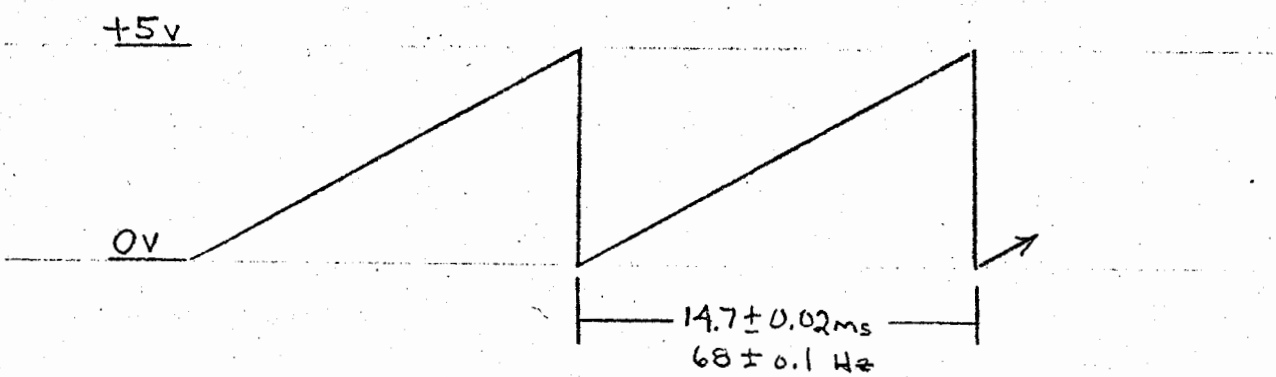



FIGURE 2.



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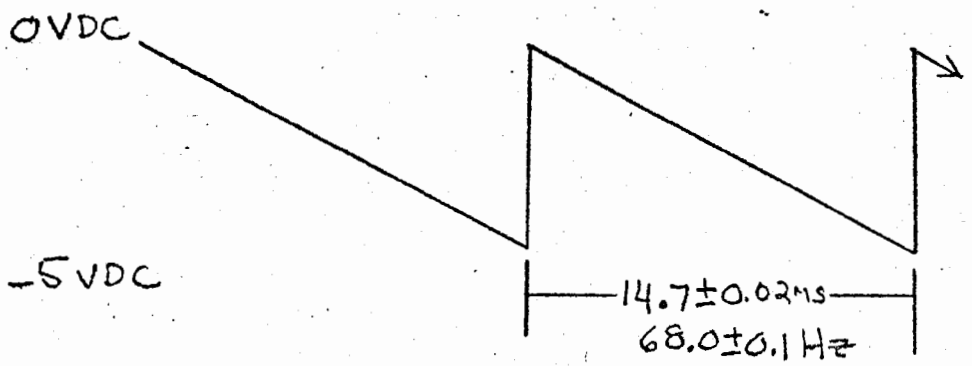



FIGURE 3

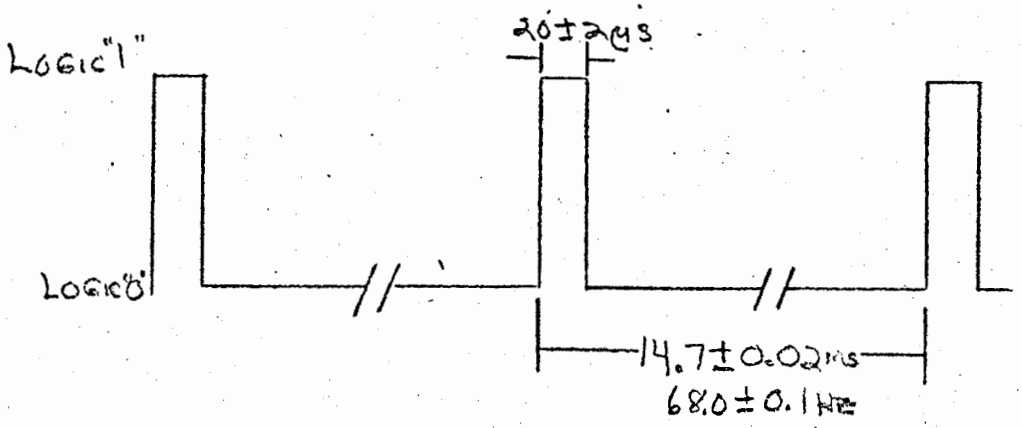



FIGURE 4

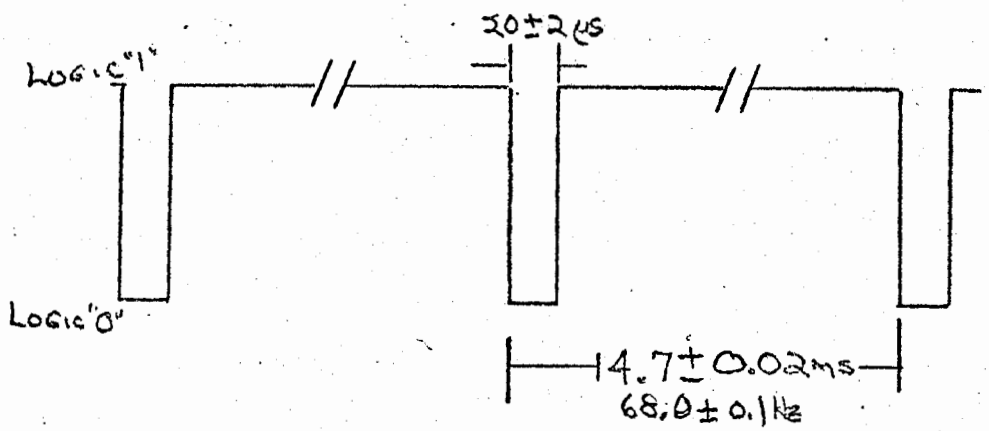



FIGURE 5



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4.2.2.6.1 Repeat the above procedure with negative voltage applied to pin 34 for negative output in column A for the same corresponding frequency.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745246.

6. NOTES

6.1 The notes specified in the general specification MIL-G-48411 are applicable to this specification.

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